

Andi Zhang

PH.D. CANDIDATE IN COMPUTER SCIENCE

Trinity College, Cambridge, CB2 1TQ, United Kingdom

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Education

University of Cambridge

PH.D. IN COMPUTER SCIENCE

- Supervised by Dr. Damon Wischik.

Cambridge, United Kingdom

Oct. 2019 - Jan. 2024 (expected)

MAST. IN MATHEMATICAL STATISTICS

Oct. 2018 - Jun. 2019

- Mathematical Tripos Part iii. Relevant modules:
Statistical Learning in Practise, Convex Optimization, Astrostatistics
Bayesian Modelling and Computation, Modern Statistical Methods

M.PHIL. IN ADVANCED COMPUTER SCIENCE

Oct. 2017 - Jun. 2018

- Relevant modules:
Machine Learning for Natural Language Processing, Algebraic Path
Deep Learning for Natural Language Processing, Advanced Functional Programming
Probabilistic Machine Learning

University of Manchester

B.SC. (HONS) IN COMPUTER SCIENCE AND MATHEMATICS

Manchester, United Kingdom

Sept. 2014 - Jun. 2017

- Relevant modules:
Machine Learning, Quantum Computing, Compilers
Convex Optimization, Probability, Statistical Methods
Functional Analysis, Topology, Number Theory

Experience

Kirin AI group, Huawei

RESEARCH INTERN (PART-TIME) | MENTOR: MR. ARNAU RAVENTOS

- Working on video frame interpolation and AI global illumination.

Cambridge, United Kingdom

Nov. 2021 - Oct. 2022

Noah's Ark Lab, Huawei

RESEARCH INTERN | MENTOR: DR. YITONG SUN

- Worked on deep learning uncertainty, out-of-distribution detection and lossless compression.

Shenzhen / Beijing, China

Sept. 2020 - Sept. 2021

Trinity College, University of Cambridge

COURSE SUPERVISOR (TEACHING ASSISTANT)

- Supervised IA Algorithms (2020 Lent), IA Machine Learning (2020 Lent),
IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas),
and IB Data Science (2022 Michaelmas)

Cambridge, United Kingdom

Oct. 2019 - Jun. 2021

MRC Biostatistics Unit, University of Cambridge

RESEARCH ASSISTANT | SUPERVISOR: DR. SOFIA VILLAR AND DR. DAVID ROBERTSON

- Built a simulation system for multi-armed bandit problem (MABP) in clinical trial.
- Presented a new test statistic for binary response MABP.

Cambridge, United Kingdom

Jul. 2019 - Sept. 2019

Tencent (WeChat AI)

R & D INTERN | MENTOR: DR. JIE ZHOU

- Concentrated on reinforcement learning and evolution strategy.

Beijing, China

Jul. 2018 - Sept. 2018

Amazon (Alexa)

APPLIED SCIENTIST INTERN | MENTOR: DR. EMILIO MONTI

- Concentrated on Neural Machine Translation (Sequence to sequence model).

Cambridge, United Kingdom

Jun. 2017 - Sept. 2017

- Assisted in implementing general stance controller for the 3D printable robotic creatures.
- Helped reproduce the prototype animation system.

Publications

Falsehoods that ML researchers believe about OOD detection

Andi Zhang, Damon Wischik

In ML Safety Workshop, 36th Conference on Neural Information Processing Systems (NeurIPS 2022)

Out-of-Distribution Detection with Class Ratio Estimation

Mingtian Zhang*, Andi Zhang*, Tim Z. Xiao, Yitong Sun, Steven McDonagh

<https://arxiv.org/abs/2206.03955>

On the Out-of-distribution Generalization of Probabilistic Image Modelling

Mingtian Zhang*, Andi Zhang*, Steven McDonagh

In Neural Information Processing Systems annual meeting (NeurIPS), 2021

Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep

Ran Guan, Andi Zhang, Mengchao Li, Yongliang Wang

In International Conference on Indoor Positioning and Indoor Navigation (IPIN), 2021

Towards dynamic and scalable active learning with neural architecture adaption for object detection

Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, Andi Zhang, Wei Zhang, Hongtao Lu, Chunjing Xu

In The British Machine Vision Conference (BMVC), 2021

Solipsistic Reinforcement Learning

Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, Andi Zhang, David Barber

In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations (ICLR), 2021

Honors & Awards

The Williams/Kilburn Medal for outstanding final year student

Manchester, United Kingdom

School of Computer Science, University of Manchester

Jul. 2017

The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.

University awards for outstanding academic achievement

University of Manchester

The University will consider the top 0.5% of undergraduates (nominated by each School) for this award.

Professors' Prize for the best third year student (joint honor)

School of Computer Science, University of Manchester

Prize for the Top 1 student.

Professors' Prize for the best second year student

School of Computer Science, University of Manchester

Sept. 2016

Prize for the Top 1 student.

Michael Jealous memorial prize for best first year student

School of Computer Science, University of Manchester

Sept. 2015

Prize for the Top 1 student.